



2 Real Analysis 2023

Tutorial 11

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Images and preimages

Problem 44 Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \sin x$. Determine the following images and preimages:

$$f\left(\left[0, \frac{\pi}{2}\right]\right), f\left(\left\{\frac{\pi}{4}\right\}\right), f(\mathbb{R}), f^{-1}(\{0\}), f^{-1}([0, 1]), f^{-1}((2, 3)), f^{-1}(\mathbb{R}).$$

Problem 45 Let X, Y be nonempty sets and $f : X \rightarrow Y$. Prove the following statements.

- (a) $f(A_1 \cup A_2) = f(A_1) \cup f(A_2)$, $A_1, A_2 \subseteq X$.
- (b) $f^{-1}(B_1 \cup B_2) = f^{-1}(B_1) \cup f^{-1}(B_2)$, $B_1, B_2 \subseteq Y$.
- (c) $f(A_1 \cap A_2) \subseteq f(A_1) \cap f(A_2)$, $A_1, A_2 \subseteq X$.
- (d) $f^{-1}(B_1 \cap B_2) = f^{-1}(B_1) \cap f^{-1}(B_2)$, $B_1, B_2 \subseteq Y$.
- (e) $A \subseteq f^{-1}(f(A))$, $A \subseteq X$.
- (f) $f(f^{-1}(B)) \subseteq B$, $B \subseteq Y$.

Use the function $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$, to show that the sign $\subseteq$ in parts (c), (e) and (f) cannot be replaced by $=$.

Continuous functions

Problem 46 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be continuous at point $x_0 \in \mathbb{R}$. Show that if $f(x_0) > 0$, then there exists $\delta > 0$ such that $f(x) > 0$ for all $x \in (x_0 - \delta, x_0 + \delta)$.

Problem 47 Suppose that two functions $f, g : \mathbb{R} \rightarrow \mathbb{R}$ are both continuous at point $x_0 \in \mathbb{R}$. Prove by definition that their sum $f + g$ is continuous at x_0 .

Problem 48 Give an example of two functions $f, g : \mathbb{R} \rightarrow \mathbb{R}$ are both discontinuous at a certain point $x_0 \in \mathbb{R}$, but both the sum $f + g$ and the product fg are continuous at x_0 .

Problem 49 Consider the function

$$f(x) = \begin{cases} 2x, & x \in \mathbb{Q}, \\ x + 3, & x \in \mathbb{R} \setminus \mathbb{Q}. \end{cases}$$

Determine all points $x \in \mathbb{R}$ where f is continuous.

Uniform continuity

Problem 50 Consider the function $f(x) = \frac{1}{1-x}$, $x \in \mathbb{R} \setminus \{1\}$.

- (a) Construct two sequences $(x_n)_{n=1}^\infty, (y_n)_{n=1}^\infty$ in $(0, 1)$ such that $\lim_{n \rightarrow \infty} (x_n - y_n) = 0$, but $|f(x_n) - f(y_n)| = 1$ for all $n \in \mathbb{N}$. Deduce that f is not uniformly continuous on $(0, 1)$.
- (b) Show that f is uniformly continuous on $\left[0, \frac{1}{2}\right]$.